Draft: Pessimistic and PB-GT Q-Learning under Adversarial State Perturbations

Adithya Venkatesh - Prof. Zizhan Zheng - July 2026 (draft for feedback)

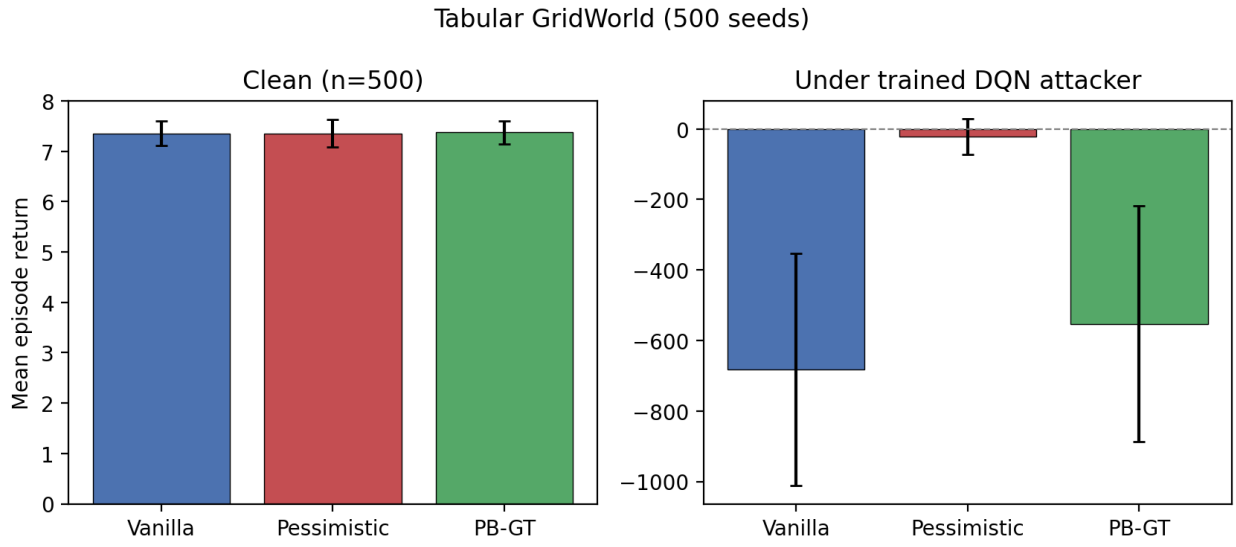
Abstract

We study belief-enriched pessimistic Q-learning (Sun & Zheng, ICLR 2024) and a Pessimistic-Behavior / Greedy-Target (PB-GT) variant that uses maximin behavior during training but greedy Double-DQN targets and greedy inference at test time. On tabular GridWorld (500 seeds), pessimistic Q-learning dominates under trained attackers (-21.8 +/- 50.0 vs -552.4 +/- 335.2 PB-GT and -682.1 +/- 329.0 vanilla), while PB-GT is ~11x faster than maximin at attack-time inference. On CartPole-v1 with Double DQN (5 seeds, vanilla best-checkpoint restore), vanilla converges reliably (499.1 +/- 1.9, 5/5 solved), but BP-style pessimistic and PB-GT show high seed variance and do not yet reproduce the GridWorld robustness ordering. We treat CartPole as preliminary and outline next steps: continuous GridWorld comparison to BP-DQN, CAR-DQN (Li et al., 2024) as a baseline, then Atari (Pong/Freeway).

1. GridWorld results (500 seeds) and interpretation

Method	Clean	Under trained attacker	Attack infer. (ms)
Vanilla	7.35 +/- 0.24	-682.1 +/- 329.0	0.008
Pessimistic	7.35 +/- 0.27	-21.8 +/- 50.0	0.093
PB-GT	7.37 +/- 0.23	-552.4 +/- 335.2	0.008

Interpretation. Clean returns are essentially identical (~7.35). Under attack, the ordering is pessimistic >> PB-GT > vanilla. Pessimistic wins over PB-GT and vanilla in >96% of seeds; PB-GT beats vanilla in ~61% of seeds. This supports the paper's claim that maximin belief-restricted policies are highly robust in discrete settings, while PB-GT is a faster approximate alternative that still improves over vanilla on average. The main cost is inference: pessimistic maximin is ~11x slower than greedy under attack.

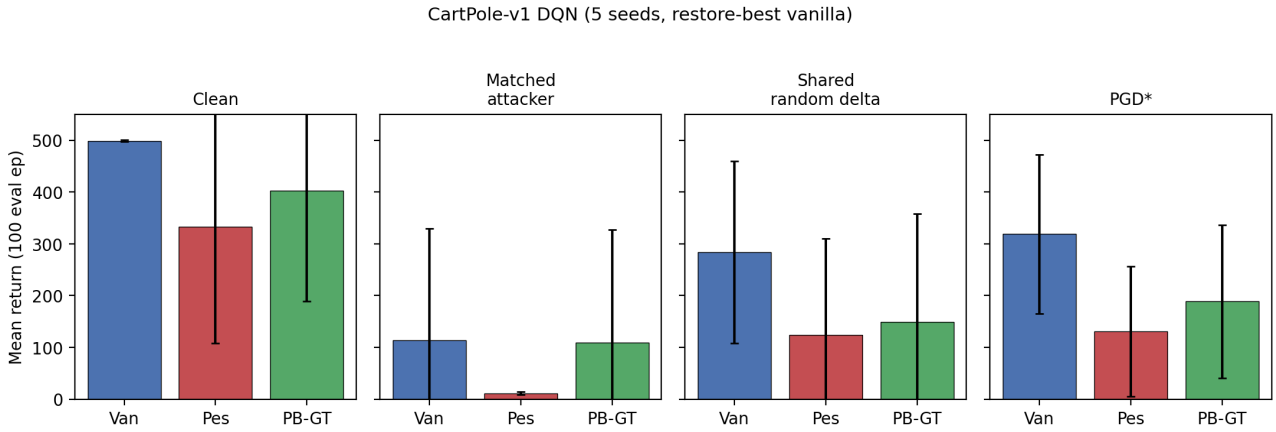


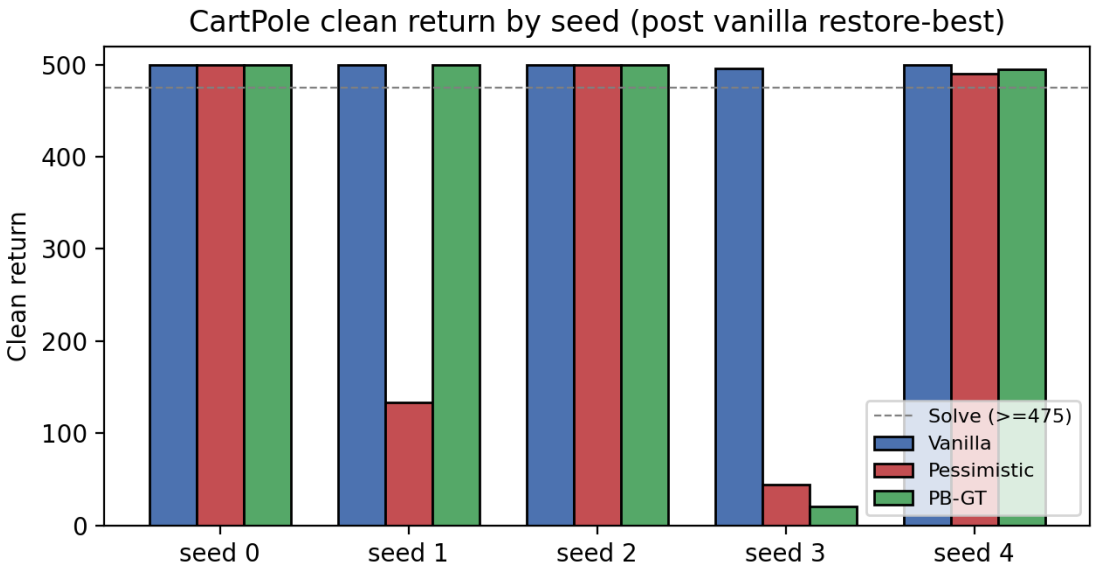
2. CartPole DQN results (5 seeds, restore-best vanilla)

Metric	Vanilla	Pessimistic (BP-style)	PB-GT
Clean	499 +/- 2	334 +/- 226	403 +/- 214
Matched attacker	113 +/- 216	11 +/- 3	109 +/- 219
Shared random delta	284 +/- 176	125 +/- 185	150 +/- 209
PGD* (32-cand.)	319 +/- 154	131 +/- 126	189 +/- 148

Interpretation. After adding mid-training eval + best-checkpoint restore for vanilla (same idea as our pessimistic v2 stabilizers), vanilla clean converges on all 5 seeds (mean 499.1). This fixes the earlier seed-2 collapse (clean 107). However, BP-style pessimistic and PB-GT are no longer stable on clean (pes 334 +/-

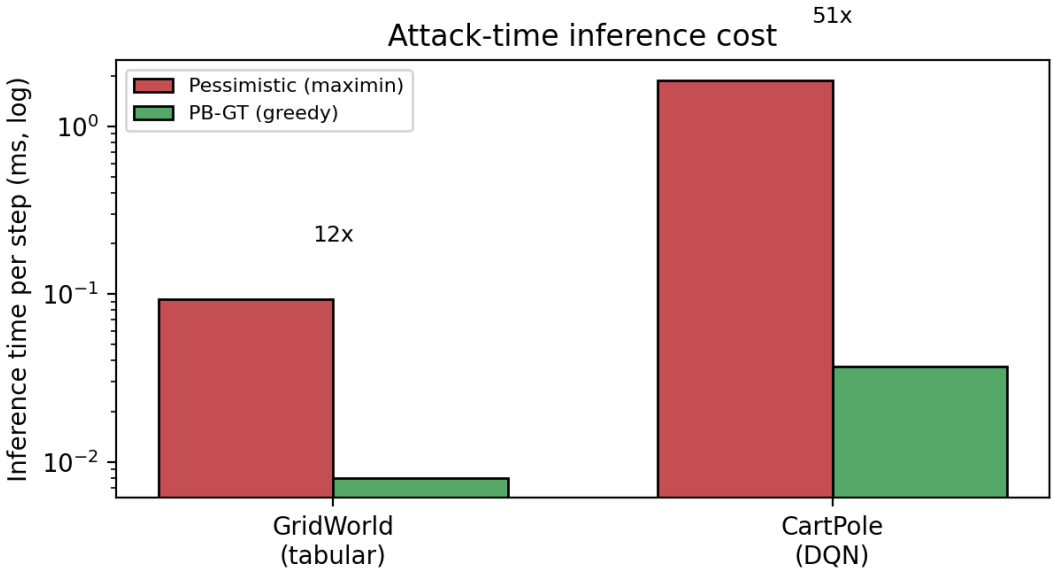
226; PB-GT 403 +/- 214), and under matched attackers pessimistic returns ~11 on every seed. We do NOT claim GridWorld-style robustness on CartPole. Likely causes: (i) fine-tune hyperparams were tuned before vanilla restore and need retuning; (ii) simplified GRU belief != PF-RNN BP-DQN; (iii) matched attackers are retrained vs each defender; (iv) n=5 is small. We keep CartPole as a preliminary DQN port and move the main deep comparison to continuous GridWorld (Xiaolin's Table 1a setting).





3. Inference efficiency (motivation for PB-GT)

Environment	Pessimistic (ms)	PB-GT (ms)	Slowdown
GridWorld attack	0.093	0.008	11.6x
CartPole matched	1.88	0.037	51x



4. Motivating PB-GT DQN (and how to improve it)

Problem. Full pessimistic / BP-DQN defense uses maximin over a belief set at every decision. That is robust in GridWorld but slow (~11-50x vs greedy) and hard to stabilize in deep RL. Many deployments need low-latency inference under moderate perturbations, not only worst-case maximin at every step.

PB-GT idea. Keep pessimistic (maximin) BEHAVIOR while collecting data under perturbed observations, so the replay buffer reflects robust exploration, but use a standard greedy Double-DQN TARGET on the true next state and a greedy TEST policy. This is a deliberate behavior-target mismatch (our GridWorld Eq. 5.1-5.2), not 'off-policy DQN' in the classical sense (all three methods are off-policy).

Why it can help. (1) Inference cost matches vanilla. (2) Training still sees adversarial observations and maximin actions, which can improve robustness over vanilla without paying maximin cost at test time. (3) On GridWorld, PB-GT sits between vanilla and pessimistic under attack while being as fast as vanilla.

Opportunities to improve PB-GT (next experiments). (A) Hybrid inference: use greedy by default, switch to maximin when belief entropy or Q-gap indicates high uncertainty. (B) Soft maximin / temperature: replace hard min_m with softmin to reduce over-pessimism during fine-tune. (C) Separate Q-heads: train a robust head with pessimistic targets and a fast head with greedy targets; distill or ensemble. (D) Retune after strong vanilla: shorter/longer fine-tune, unfreeze more layers, align kappa and PGD with Algorithm 4. (E) Evaluate on continuous GridWorld against BP-DQN and CAR-DQN under identical PGD / MinBest / PA-AD protocols before claiming deep-RL gains.

5. Next experimental steps (advisor roadmap)

1. Continuous GridWorld (priority). Use Xiaolin's environment and BP-DQN pipeline (github.com/SliencerX/Belief-enriched-robust-Q-learning; config Grid_continous_ours.json; train.py / test_gridmaze.py). First reproduce vanilla DQN and BP-DQN Table 1a numbers, then add PB-GT, then increase seeds for a rigorous comparison.

2. CAR-DQN baseline (Li et al., arXiv:2402.02165 / ICML). Consistent Adversarial Robust DQN minimizes a surrogate of Bellman Infinity-error under the Consistency Assumption of Policy. Code: github.com/leoranlmia/CAR-DQN. Include as an additional robust DQN baseline on continuous GridWorld (and later Atari) alongside SA-DQN / WocaR-DQN where feasible.

3. Then Pong / Freeway + DP-DQN, with GPU workstation support (Yibin).

4. CartPole: optional retune of pessimistic / PB-GT after vanilla restore; treat as appendix / preliminary unless continuous GridWorld confirms the narrative.

6. Limitations (current draft)

CartPole pessimistic is BP-style (simplified GRU particles, kappa=20), not full PF-RNN Algorithm 4. PGD* is defender-specific 32-candidate search, not paper PGD (eta=10). Matched attackers are not a shared threat. n=5 on CartPole. Continuous GridWorld and CAR-DQN results are not yet included.

References

[1] X. Sun and Z. Zheng. Belief-Enriched Pessimistic Q-Learning Against Adversarial State Perturbations. ICLR 2024.

[2] H. Li et al. Towards Optimal Adversarial Robust Q-learning with Bellman Infinity-error (CAR-DQN). arXiv:2402.02165, 2024.

[3] V. Mnih et al. Human-level control through deep reinforcement learning. Nature, 2015.

[4] Code: SliencerX/Belief-enriched-robust-Q-learning; leoranlmia/CAR-DQN.

Appendix: reproducibility

GridWorld: main_with_vanilla.py, gridworld_experiments_500runs.csv. CartPole: main_cartpole_dqn.py, cartpole_belief.py, cartpole_experiments_restore_5runs.csv. Methods writeup: arxiv_prep_algorithms_metrics.pdf. Regenerate this draft: python3 generate_paper_pdf.py